

YITIAN WANG

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SUMMARY

Ph.D. scholar in Electrical Engineering specializing in thermal transport and lattice dynamics in functional oxides and solid-state ionic conductors. Experienced in crystal and thin-film growth, vacuum deposition, characterization, and data-driven modeling using Python and MatLab. Published 5 first-author works in leading journals and skilled at collaborating with national labs and cross-disciplinary device and materials teams. Extensive teaching background, including tutoring since undergraduate years and three terms of teaching assistant experience during Ph.D. studies.

EDUCATION

University of California, Riverside, CA *Mar. 2021 - Jun. 2026 (expected)*
Ph.D. in Electrical and Computer Engineering
Awards: MRS 2023 Spring Highly Commended Student Talk Award
Dissertation Completion Fellowship Award

Columbia University in the City of New York, NY *Aug. 2019 - Feb. 2021*
Master of Science, Material Science and Engineering

University of California, Berkeley, CA *Jul. 2017 - Aug. 2017*
Summer Session, Physics

Beijing Normal University, CN *Sep. 2015 - Jun. 2019*
Bachelor of Science, Physics
Awards: 'Jingshi' Scholarship (2016 - 2018)

SKILLS

Software / Programming

- Python (advanced), MatLab (advanced), SolidWorks (intermediate)
- GSAS-II (advanced), Mantid (advanced), ImageJ (intermediate)

Experimental Techniques

- Floating-Zone Crystal Growth (advanced), Physical Vapor Deposition (intermediate), Chemical Vapor Deposition (intermediate)
- Scanning Electron Microscopy (advanced), X-Ray Diffraction (advanced), Inelastic Neutron Scattering (advanced), Raman Spectroscopy (intermediate), Fourier-Transform Infrared Spectroscopy (intermediate)
- Physical Property Measurement System Measurements (advanced), Differential Scanning Calorimetry (intermediate), Thermogravimetric Analysis (intermediate)
- Prototype Battery Assembly (advanced), Electrochemical Impedance Spectroscopy (intermediate)

RESEARCH EXPERIENCE

Graduate Research Assistant

University of California

Jun. 2021 - Jun. 2025

Riverside, CA

- Studied thermal transport and ionic mobility in solid-state electrolytes and complex oxides.
- Grew single-crystal oxide Li-ion conductors using a floating-zone furnace.
- Characterized structure and transport with XRD, transport measurements, and electrochemical testing.
- Built a two-channel thermal-conductivity fitting model in MatLab.

Experiment Lead

Oak Ridge National Laboratory

Jan. 2022 - Jul. 2023

Oak Ridge, TN

- Led inelastic neutron-scattering studies on LLZTO using TAX at HFIR and ARCS at SNS.
- Measured phonon spectra of single-crystal solid electrolytes.
- Processed scattering datasets and automated analysis workflows in Mantid and Python.

Student Researcher

Columbia University

Sep. 2019 - Jan. 2021

New York, NY

- Developed composite lithium-ion battery separators with γ - C_3N_4 and PVDF using solution processing.
- Fabricated prototype coin cells and measured impedance and galvanostatic cycling.
- Ran first-principles simulations in Quantum ESPRESSO on defect-engineered olivine structures.

Student Researcher

Beijing Normal University

Sep. 2018 - May 2019

Beijing, CN

- Designed low-temperature uniaxial-strain devices in SolidWorks.
- Characterized detwinning effects in $BaFe_2As_2$ with PPMS.

SELECTED PUBLICATIONS

- Glass-like thermal transport in polycrystalline perovskite lithium-ion conductor $Li_{3/8}Sr_{7/16}Hf_{1/4}Ta_{3/4}O_3$
 - *Chemical Communications*, 2025 [\[Link\]](#)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Low Thermal Conductivity and Lattice Anharmonicity of NaSICON-type Solid Electrolyte $Na_3Zr_2Si_2PO_{12}$
 - *Tungsten*, 2025 [\[Link\]](#)
 - **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen
- Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte: Linking lattice and ionic dynamics with thermal transport
 - *PRX Energy*, 2025 [\[Link\]](#)
 - **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, Y. Li, H. Li, J. He, Y. Xu, S. Guo, Q. Cai, D. L. Abernathy, T. Williams, K. V. Kravchyk, M. V. Kovalenko, G. K. H. Madsen, C. Li, X. Chen
- Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$

- *Journal of Materials Chemistry A*, 2024 [\[Link\]](#)
- **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen
- Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $\text{Sr}_{14-x}\text{Y}_x\text{Cu}_{24}\text{O}_{41}$
 - *Journal of Applied Physics*, 2024 [\[Link\]](#)
 - S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen
- Size-dependent magnon thermal transport in a nanostructured quantum magnet
 - *Cell Reports Physical Science*, 2024 [\[Link\]](#)
 - S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen
- Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe_4
 - *Chemistry of Materials*, 2022 [\[Link\]](#)
 - Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, A. A. Rezaie, V. Carta, K. N. Bozhilov, D. Luong, B. P. T. Fokwa, F. Kargar, A. A. Balandin, X. Chen
- Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors
 - *Tungsten*, 2022 [\[Link\]](#)
 - **Y. Wang**, X. Chen